

Lab Session 1: Network Reconnaissance and Host Discovery using Nmap

Date: 26 May 2026

Time: 19:00

Duration: 50 mins

Scenario:

You are a Jr SOC analyst at a fintech firm. Your manager asks you to map the internal test network (192.168.56.105) to identify exposed services before the upcoming audit.

Objective:

Use Nmap to discover hosts on a network, enumerate open ports, identify running services, detect OS fingerprints, and understand what an attacker can see before attempting exploitation

Environment:

Kali IP: 192.168.56.1/24

Target IP: 192.168.56.105

Subnet: 192.168.56.0/24

Tools Used: Nmap Terminal

Virtual Machines Running: Metasploitable 2 VM on Virtual Box

Commands, step by step:

1. Sudo nmap -sn 192.168.56.105
2. Sudo nmap -p- 192.168.56.105
3. Sudo nmap -sV 192.168.56.105

4. Sudo nmap -A 192.168.56.105
5. Sudo nmap -O 192.168.56.105

What do they do? (Respectively)

-sn is a ping sweep for host discovering, it finds all live or online hosts without scanning any open ports

-p- is a full port scan , shows all open ports on the host

-sV is a service and version detection command, it identifies what software and versions are running on each open port

-O identifies the target's operating system

-A is an Aggressive All in one scan, shows OS, version,scripts and traceroutes

Findings:

- ❖ **sudo nmap -sn 192.168.56.105** the target responded instantly, confirming the VirtualBox (VB) network adapter is correctly configured. The host is live and reachable within the private 192.168.56.0/24 subnet.
- ❖ **sudo nmap -p- 192.168.56.105** revealed 30 open ports. Ftp, ssh and http were among them, these are ports that are usually targeted by attackers for network exploit and reconnaissance.
- ❖ **sudo nmap -sV 192.168.56.105** Successfully isolated 1 unique active IP. The scan accurately mapped the banners of the 30 services. The OS detection engine gathered enough distinct TCP sequence metrics to provide a fingerprint of the underlying operating system.
- ❖ **sudo nmap -O 192.168.56.105** Nmap sent a specialized series of TCP and UDP packets to the target's open and closed ports, measuring subtle differences in how the target's network stack responds. This provided a high-confidence

identification of the underlying operating system type and kernel version running on the VirtualBox VM.

- ❖ *sudo nmap -A 192.168.56.105* The aggressive scan executed as intended, forcing the Nmap Scripting Engine to run default security scripts against the 30 open ports. This collected deeper contextual data, such as supported SSH, HTTP server headers, and potential low-hanging vulnerabilities.
- ❖ Results saved however protocol negotiation failed find out whats up with that

Made a mistake : In future make sure your commands are specific to the target and not basic eg : *sudo nmap --script vuln* without an IP will not yield results like you saw, however *sudo nmap --script vuln 192.168.56.105* did the vulnerability scan and gave the results you were looking for and /webdav which was flagged as a potentially interesting folder

Post Lab:

Protocol Negotiation failed due to miscommunication, update nmap and run it again in next lab before you move on to next lab

What's Next?

Vulnerability Scanning and Web App Recon using Nikto and Gobuster